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Intelligent Data Access: How AI-Powered Search Transforms Subsurface Knowledge Management

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Abstract

The oil and gas sectors face significant challenges in managing vast, disparate geoscience datasets. This project addresses these challenges by implementing a centralized solution for subsurface data and documents, providing users with rapid access to data through web-based tools. Generative Artificial Intelligence (AI) offers conversational natural language queries across unstructured data and augments visualization with summaries of complex subsurface data environments. The solution integrates a unified subsurface data lake with standardized metadata tagging and a custom-trained natural language processing (NLP) engine specifically fine-tuned for geoscience terminology. Secure API connections provide comprehensive access to well databases, seismic repositories, and technical reports.

Early adoption has demonstrated transformative benefits, including a 70% reduction in time spent searching for information and a 60% decrease in technical data management help desk requests related to data location. The intelligent data access interface lowers the barrier to accessing and understanding subsurface knowledge, empowering teams to uncover insights from previously disconnected information. By converting complex data retrieval into a simple, conversational process, this project demonstrates how AI-powered search can fundamentally change how geoscientists interact with technical data, turning raw information into actionable knowledge at the speed of conversation. This approach aligns with industry trends highlighted by firms such as Gartner and McKinsey, which emphasize the importance of breaking down data silos and using generative AI to automate knowledge work and enhance decision-making.

Introduction

The geoscience domain is inherently data-rich, characterized by an enormous volume of information collected from various datasets, including seismic surveys, well logs, core analyses, and technical reports. This data is critical for making informed decisions regarding exploration, development, and production. However, the sheer scale and complexity of these datasets have created a significant and persistent problem: data accessibility and effective knowledge management. Historically, this information has been stored in a fragmented landscape of siloed databases, legacy systems, and unstructured documents. This fragmentation, along with inconsistent data formats and a lack of standardized metadata, means that geoscientists often spend a disproportionate amount of their time hunting for information rather than analyzing it. A study by the Society of Petroleum Engineers (SPE) found that geoscientists can spend up to 70% of their time on data-related tasks rather than on core analytical work, a staggering inefficiency that directly impacts a company's ability to innovate and compete. This inefficiency translates into significant economic and strategic consequences, including delayed exploration projects, suboptimal well placements, and missed opportunities to extract value from existing assets.

Traditional knowledge management systems, such as Structured Query Language (SQL) databases and Enterprise Content Management (ECM) platforms, require specialized technical expertise and a precise understanding of data schemas. While effective for specific tasks, these tools create a barrier between the domain expert—the geoscientist—and the data itself. A geoscientist seeking to answer a complex, multi-parameter question often needs to navigate multiple systems, perform manual data joints, and even enlist the support of technical data management teams. This cumbersome process hinders productivity, stifles innovation, and prevents the discovery of meaningful connections that exist across disparate datasets. This is particularly true for "dark data" - the vast amounts of unstructured or poorly documented data that are effectively invisible to traditional search methods. The inability to easily access and correlate this data can lead to suboptimal decisions, project delays, and the repeated collection of data that already exists, all of which represent tangible costs to the business.

The advent of powerful conversational AI and Large Language Models (LLMs) presents a unique opportunity to address these challenges. These technologies have demonstrated an ability to understand and process natural language with remarkable accuracy, opening the door for more intuitive human-data interaction. This paper presents a novel solution that leverages these advancements to create a user-friendly, conversational AI interface for subsurface data. Our project, titled the Advanced Subsurface Data Platform, is designed to dismantle the barriers to data access by allowing users to retrieve information using plain English, effectively transforming the knowledge-seeking process from a technical chore into a natural conversation.

The primary objective of this paper is to detail the architecture and implementation of the project. We will present a comprehensive overview of how this system integrates a centralized data lake, standardized metadata, and a custom-trained NLP engine to deliver accurate and rapid responses to geoscience queries. Furthermore, we will present evidence from early adoption and pilot programs, showcasing how the tool significantly reduces information retrieval time, decreases reliance on data management support, and empowers users to uncover valuable insights with unprecedented ease. This research serves as a foundational case study for how AI-powered search can be applied to complex technical domains, fundamentally changing the paradigm of data interaction.

The Subsurface Knowledge Challenge: A Review of Current and Historical Practices

For decades, the energy industry has grappled with the exponential growth of digital geoscience data. The scale of this data is immense, ranging from petabytes of seismic data to terabytes of well logs and millions of pages of technical reports. The challenge is not just in storage, but in retrieval and synthesis. To understand the current pain points, it is essential to review the historical evolution of subsurface data management first.

The Historical Context of Data Management

The history of geoscience data management can be broadly categorized into three distinct eras: the Paper Era, the Digital Era, and the Integrated Digital Era. During the first Paper Era, all data—maps, well logs, reports—were physical documents. Data retrieval was a laborious manual process of searching through filing cabinets and map rooms. The second was the Digital Silo Era, where data was digitized but stored in departmental, application-specific databases. While this was a significant step forward, it created the fragmentation and "silo" problem we see today. Geologists' data was stored in a geological database, geophysicists' data was stored in a seismic database, and engineers' data was stored in a reservoir engineering system, with little to no interoperability. A geophysicist might need to access well log data from a separate department's system. This process requires manual data exports, format conversions, and often physical transfer via external storage devices. This era was defined by a reliance on specialized software and a lack of a unified view of the subsurface. The third and current era is the transition towards the Integrated Digital Era, where the industry is moving towards a unified data ecosystem. Our project represents a key part of this transition, providing the final piece of the puzzle: an intelligent access layer.

The Problem of Data Fragmentation and Silos

The siloed nature of geoscience data is a direct result of the specialized workflows and software used in the field. For example, a geologist working on a prospect might use geological modeling software that stores its data in a proprietary format. A geophysicist might use another interpretation software with its own unique database schema. When the geologist needs to validate their model with seismic data, they must manually export, convert formats, and import the data, a process that is not only time-consuming but also prone to error and data loss. This absence of a common data language is a significant barrier to cross-disciplinary collaboration.

Furthermore, a significant amount of valuable information is often locked away in unstructured formats, such as PDF documents, scanned images, and legacy spreadsheets, frequently referred to as "dark data." This is data that exists within the organization's network but is not easily discoverable or analyzable. A petrophysicist might spend weeks searching for a specific report in a document repository because they only remember a keyword or the author's name, not the exact file path or name; this often happens when a previous interpreter leaves the company without proper handover documentation. This makes a large portion of a company's intellectual property effectively undiscoverable without a dedicated manual search.

Limitations of Traditional Knowledge Management Methods

Traditional methods for accessing this data typically fall into one of three categories, each with its own set of limitations:

- **Direct Database Access:** Requires a user to have a deep understanding of database schemas and be proficient in a query language like SQL. A geoscientist, for example, would need to know the specific table names, column names, and syntax to query for all well logs with a specific lithology. This is not a scalable solution for an entire organization, as it creates a heavy reliance on a small number of data experts.
- **Specialized Software Interfaces:** Applications like seismic interpretation software or reservoir modeling tools provide access to specific data types but are not designed for cross-dataset queries or natural language interaction. A geophysicist cannot, for example, use their seismic interpretation tool to query for well logs with a certain petrophysical property in a completely different database.
- **Manual Search and Technical Data Management Support:** For complex queries or for locating unstructured data, users often resort to manual searches of network drives or submit a help desk ticket to a data management team. This process is slow and creates an organizational bottleneck. A technical data management team can be overwhelmed with simple requests for data location, diverting their expertise from more strategic data management tasks.

These limitations underscore a critical need for a new paradigm—one that lowers the technical barrier to entry, promotes cross-disciplinary data discovery, and makes information instantly accessible. The solution must be capable of understanding the user's intent, regardless of the data's underlying structure, and provide a unified view across the entire data landscape. The proposed Advanced Subsurface Data Platform tool is a direct response to this need, leveraging modern AI to bridge the gap between human language and complex, fragmented data.

Our Organization's Data Challenges

While the issues outlined above are common across the industry, our organization faced several unique and compounding challenges amplified by our corporate history and growth trajectory in Malaysia:

- **Legacy data across multiple systems:** Key subsurface data was spread across different repositories and applications, with some still in outdated formats and lacking standardized metadata. This challenge was further compounded following PTTEP's expansion in Malaysia.
- **Time-consuming data searc**h: Geoscientists frequently spend hours or even days locating and validating data prior to analysis. The integration of PTTEP Malaysia operations increased volume and diversity of data, sometimes resulting in duplicate records and inconsistent naming conventions, which slowed project timelines.
- **Limited access to unstructured data:** A significant portion of valuable information exists in unstructured formats (e.g., PDF reports, scanned images, excel files), dispersed across many folders and multiple versions. Data often lacks proper metadata or indexing, making retrieval even more challenging.
- **Dependency on technical data management teams:** Complex data retrieval often requires specialized data management support, creating bottlenecks and delaying decision-making. The need to reconcile legacy and new datasets added further complexity to data access.

These challenges highlight the necessity for a modern platform that simplifies access, reduces manual dependencies, and unlocks the full potential of both structured and unstructured datasets, ensuring seamless integration across legacy and newly acquired data sources.

Methodology and System Architecture

The Advanced Subsurface Data Platform is built on a robust, four-layered architecture designed for scalability, security, and performance. The system integrates a unified data management layer, an AI-powered conversational engine, and secure API connections to facilitate seamless communication and interaction.

The Unified Subsurface Data Lake and Metadata Layer

The foundation of the Advanced Subsurface Data Platform is a unified subsurface data lake. This centralized repository consolidates geoscience data—including structured databases, unstructured technical reports, depth registers geological sample images and semi-structured seismic data—into a single, accessible location. A critical component of this data layer is the implementation of a standardized metadata tagging framework. This framework ensures that every data asset, regardless of its origin or format, is consistently tagged with key information. This includes, but is not limited to:

- **Geospatial metadata:** Location, coordinate system, survey boundaries.
- **Temporal metadata:** Date of acquisition, date of last modification.
- **Geological metadata:** Formation names, zone names, lithology types.
- **Technical metadata:** Data type (e.g., well log, seismic 3D), processing parameters.
- **Access metadata:** Owner, access control list, security level.

This metadata tagging is the backbone of the system. It enables the AI engine to quickly understand the context of a user's query and efficiently locate relevant data points, even when the data itself is stored in different formats or databases. Without this tagging, the AI would struggle to accurately interpret queries or retrieve data from disparate sources, resulting in slower and less reliable results.

The implementation of this data lake and metadata framework required a significant data ingestion and curation process, as shown in Figure 1. Data from legacy systems and unstructured files were extracted, transformed, and loaded into the data lake. The curated data is stored in relational and file-based datastores, enabling access through collaborative workspaces, AI-driven search, and visualization tools such as correlation and well viewers.

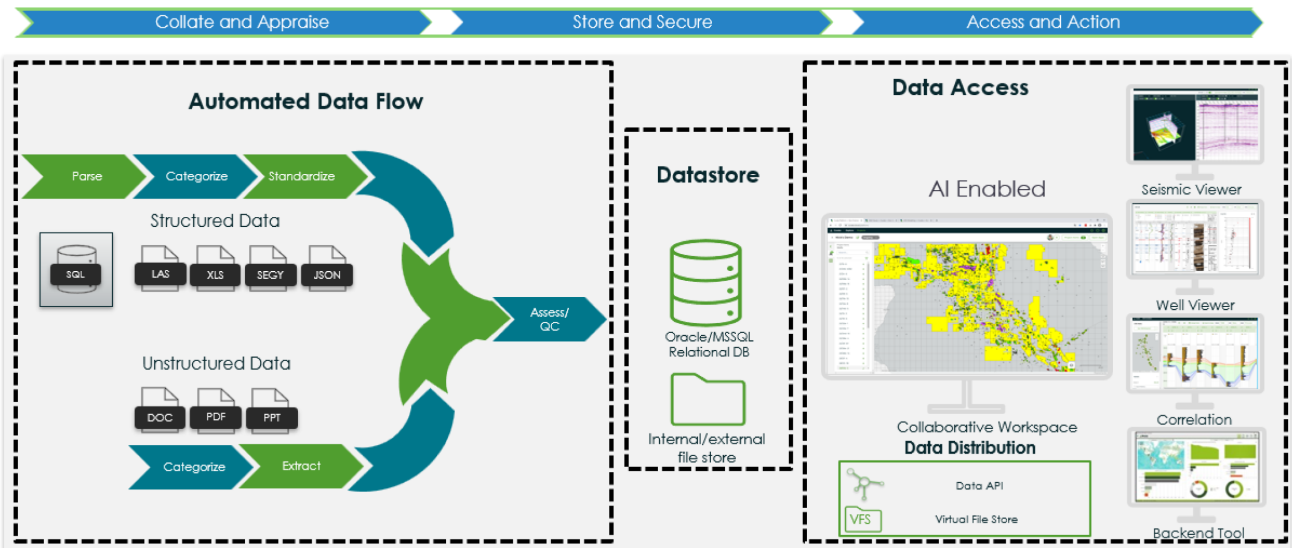

Figure 1—Advanced Subsurface Data Platform – Ingestion to Insight

Web-based Data Consumer Access & Collaboration

Traditional subsurface data management systems are typically challenging for users to navigate and identify data that is valuable for their workflows. The Advanced Subsurface Data Platform provides web-based access for the subsurface data consumer community. The tools are user focused on enabling rapid data discovery and validation of the relevance of data for their subsurface workflows. Tools enable synthesis and visualization of complex subsurface datatypes within the data platform removing the need to export data only to find it is of poor quality of little relevance. Collaboration tools allow cross-discipline review of data during data gathering, for example a Geologists, Petrophysics and Geophysicist working together to identify suitable input for a seismic inversion and quantitative interpretation project. Web based tools include:

- **Map Search:** Intuitive map search including geographic lasso tools and metadata-based filters. The map interface also includes available data matrix summaries and traditional data and document lists.
- **Visualization:** Tools to visualize a wide range of technical data including Single Well Views, Multi-Well Correlations, Maps of Well data and properties, 2D/3D Seismic viewing
- **Logical Search:** Data Explorer tool to allow data consumers to build rapid logical searches of structured database. For example, show me wells with have Vp, Vs , Density and interpreted Volume of Shale within a specified formation or depth range

The AI-Powered Search and Summary

The Advanced Subsurface Data Platform is augmented with OpenAI-powered tools to streamline user data discovery and understanding. The components of the AI toolkit include:

- **Document Ingestion:** Documents are automatically attached to well or seismic data objects in a streamlined ingestion process
- **Document Vectorization:** To enable the ChatAI workflow, documents are broken into chunks of text, and a vectorized database of the documents is generated using OpenAI embeddings API.
- **User Query:** The user enters a natural language query via the chat interface, and the query is also vectorized using the embeddings API.
- **Quest Response**: Similarity between the vectorized User Query and vectorized document database is assessed, and the vectorized query and similar chunks of vectorized documents are processed through the OpenAI Completions AI to generate a response for the user. Prompt engineering is used to ensure the responses are relevant to subsurface users.
- **Document Reference**: To provide assurance that the Generative AI tools are giving accurate answers, users are also provided with links to references with the original source documents from which responses are generated.

AI Inform is a series of tools to provide rapid summaries of information to data consumers. This includes:

- **Document Summary:** Users can instantly be provided with short summaries of the contents of a document. Using similar workflows to the ChatAI functionality, summaries are generated for a specific document through the web-based front end.
- **Structured Data Summaries:** Users can generate their own structured data views, for example, a detailed single well view of raw logs, petrophysical interpretation, measured discrete core data, and sample images, or a regional correlation of lithological units. AI inform allows users to generate AI summaries of specific intervals of interest, for example, around a potential reservoir pay zone. Prompt engineering is used to generate meaningful summaries of visualized data, whether it be petrophysical interpretation or drilling parameters.

Secure Data Pipeline and Data Governance

To ensure comprehensive data access, the Advanced Subsurface Data Platform utilizes a secure data pipeline that includes connections to pre-existing databases and the automated ingestion of unstructured data stored on network drives. Secure connections enable web-based tools to interact with aggregated relational databases and associated unstructured documents. A crucial aspect of architecture is rigorous data governance. All data access is governed by role-based access control (RBAC). When a user logs into the system, their permissions are authenticated, and the system ensures they can only access data for which they have explicit authorization. This prevents unauthorized users from querying sensitive information, providing a secure and compliant data environment. RBAC control is pervasive across data administrator tools, data consumer tools, AI query, and data distribution tools.

Results and Transformative Benefits

The implementation of the Advanced Subsurface Data Platform has yielded significant, quantifiable benefits during its early stages of adoption. A pilot program was conducted with a team of 15 geoscientists and engineers over three months to measure its impact. The results demonstrated a paradigm shift in how users interacted with technical data.

Quantifiable Metrics: Speed and Efficiency

The most immediate and impactful result was the significant reduction in time spent searching for information. Through a series of timed tasks and user surveys, the tool was found to provide a 70% faster information retrieval rate compared to traditional methods. For example, a query like "Show me all wells with porosity >15% and a Gamma Ray reading <50 API units in the A Formation" that would have previously required a geoscientist to manually query two separate databases and join the results was answered by the tool in under 15 seconds.

Furthermore, the pilot program tracked internal technical data management team tickets related to data requests. Prior to deployment, a significant portion of the technical data management team's requests involved locating specific files, finding the latest version of a report, or requesting data extracts. The adoption of the tool led to a 60% reduction in help desk requests for locating data. This not only frees up valuable time for the data management team but also indicates that users are now self-sufficient in their data discovery needs.

Qualitative Impact: Knowledge Discovery and Collaboration

Beyond the efficiency gains, the tool has had a transformative qualitative impact. By lowering the barrier to entry, it has enabled users to explore data in ways that were previously too time-consuming or technically challenging. The most significant qualitative benefit is the new ability to discover meaningful connections across previously disconnected datasets. For instance, a geoscientist was able to utilize the ChatAI functionality to identify all the main reservoirs of interest within a field and then employ the logic tools to pinpoint wells with data suitable for parameterizing a seismic inversion within the reservoir of interest. This type of complex, multi-parameter query, which synthesizes information from seismic and interpretation reports, well, and technical report data, would have been nearly impossible to execute in a timely manner using legacy systems.

The tool also proved to be an invaluable tool for new team members. Instead of spending weeks learning the complex data landscape and navigating various software interfaces, new geoscientists could use the AI assistant to ask questions like "What are the key geological formations in Block A?" or "Where can I find the most recent petrophysical study for the Y field?" This guided discovery process significantly shortens the onboarding time and accelerates their path to productivity. The feedback from users was overwhelmingly positive, with many describing the interface as "intuitive" and "empowering." This sentiment encapsulates the project's core value: turning centralized information into actionable knowledge at the speed of conversation.

Future Work and Limitations

While the tool has demonstrated remarkable success, several avenues for future development and acknowledged limitations remain.

Limitations

The system currently leverages traditional data management workflows to provide end users the opportunity to interrogate the structured data, and Generative AI to query unstructured data. Therefore, querying the structured data has a limitation from its reliance on the quality of the underlying metadata. If data is not consistently or accurately tagged, the user's ability to find and interpret it is diminished. While ChatAI functionality is closely coupled with the associated documents and provides users with links back to the original unstructured documents, there is currently no functionality to relate the responses back to geographic features in the platform's Map visualizations.

Future Enhancements

The current implementation is focused on information retrieval. Future work will explore expanding the tool's capabilities to include:

- **Data Visualization:** Automatically generating charts, maps, and plots in response to queries (e.g., "Show me a map of wells which encountered drilling issues in Formation X in the last 5 years").
- **Predictive Analysis:** The architecture of the platform supports the development of bespoke external workflows accessing the data through API. This provides the opportunity to integrate with machine learning models to answer predictive queries (e.g., "Predict the porosity distribution in a given area based on seismic attributes").

Conclusion

The Advanced Subsurface Data Platform project represents a significant leap forward in subsurface data management. By leveraging the power of conversational AI, we have successfully developed a system that transforms a complex and time-consuming data retrieval process into a simple and intuitive experience. The system's architecture, built upon a unified data lake, a custom-trained NLP engine, and robust security protocols, has proven its effectiveness in a real-world setting.

Early adoption metrics, including a 70% reduction in search time and a 60% decrease in help desk requests for data, provide compelling evidence of the system's transformative power. Beyond these numbers, the tool empowers geoscientists to ask more sophisticated questions, make novel connections across disparate datasets, and accelerate their path to insight. By democratizing access to subsurface knowledge while upholding rigorous data governance, the AI-powered search interface successfully lowers the barrier to entry for both new and experienced professionals. The future of geoscience data interaction will not be defined by complex database queries, but by natural language conversations, turning data hunting into knowledge asking.

References

1. Viridien. (2022). "Extracting Knowledge With NLP From Massive Geological Documents."
2. Chen, B., et al (2023). "Data Lakes: A Survey of Concepts and Architectures."
3. McKinsey & Company. (2022). "Harnessing volatility: Technology transformation in oil and gas."
4. EY. (2025). "How AI drove data optimization for oil and gas capital projects."
5. Gartner. (2025). "Generative AI: What Is It, Tools, Models, Applications, and Use Cases."
6. Number Analytics. (2025). "AI in Petroleum Data Management."